

Technology Stack Specification

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Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React (Vite), Tailwind CSS, Recharts, Framer Motion	Facilitates highly responsive component reusability, dark mode styling, and dynamic charts rendering.
2	Backend / Server-Side	FastAPI (Python 3.12), SQLAlchemy ORM	Provides high-performance asynchronous REST routing and clean object-relational mappings.
3	Database / Data Storage	SQLite	Zero-config file database perfect for single-user containment and rapid local testing.
4	Cloud / Hosting / Deployment	Docker, Docker-Compose	Enables consistent container orchestration across local and production deployment pipelines.
5	Version Control & CI/CD	Git / GitHub	Tracks codebase histories, branch staging, and project template submission integrations.
6	Third-Party APIs / Other Tools	Google Gemini API, ReportLab, OpenPyXL	Powers the advisor chat/recommendations, and compiles PDF/Excel file streams in-memory.